

Stages of the research process

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This lecture provides the student with an overview of the stages of the research process. Research is carefully planned and systematically carried out. There are choices that the researcher has to make concerning certain aspects of the research, for example data collection and analysis methods. These will be done in more detail as the module progresses. The plan for the research is presented in a research proposal. The research proposal is a meticulous document and its creation is described here.

1. The research process

The basic stages that are undertaken to carry out research is known as the *research process*. There are generic and essential stages that need to be taken in a systemic manner. A sound research process ensures high quality and plausible research as well as demonstrates scientific rigour.

An understanding of the research process is necessary to effectively carry out research and sequencing of the stages inherent in the process. The stages below are generic to most research projects:

1. Identifying the problem
2. Draft research questions, objectives, and hypotheses
3. Review relevant literature
4. Determine the research strategy
5. Data collection and sampling
6. Data analysis
7. Writing up findings

The sequence of stages is important as there is a dependency on the previous stage.

It is important to understand the term *iterative*. Iterative means going back to improve because new knowledge and better understanding has become available. Some of the stages are iterative in nature, particularly stages 2, 3 and 4.

1. Identifying the problem

Not all problems around us are suitable for a research undertaking. A research problem is one with special characteristics. A researchable problem is one that requires scientific process of enquiry and includes the collection and analysis of data.

A research problem results in a sense of experienced difficulty. It is not a straightforward problem and the solution is not straightforward. A research problem could have two or more potential solutions. If there is a single definitive solution to the problem, then it is not a researchable problem.

For a research problem, there is a difference between the *current situation* and the *ideal situation*. The reasons for this difference are either unknown or not well understood. The research problem should tell the reader about “What the problem is?”, “Where is it occurring?”, “Why is it occurring”, and “When is it occurring”. Remember that a reader might pick up the work 1000 years from today.

Once the problem is understood, a clear statement of the problem should be formulated. This is often referred to as the *research problem* and is the foundation for the rest of the research. The research problem provides details of where the problem is occurring, what its effects are and why it is important to research and find a solution. From the research problem, a *research question* and *objectives* are formulated.

In some cases, exploratory research can be undertaken prior to formulating the problem statement.

e.g. research problem adapted from Sung (2007)

“Computing is the foundation of modern society post 2000s. A proficient computing workforce is essential for maintaining the country’s leadership and competitiveness in the global economy. For this reason, the recent decline in enrolments across Computer Science (CS) departments and the decrease in student diversity pose significant challenges to the continuation of the nation’s prominent position in the global high technology arena. An immediate solution is urgently needed. For the health of the national economy, and to sustain successful and vibrant software companies like Microsoft, we must build excitement and enthusiasm for our discipline in order to attract a bright new generation of students early in their academic careers.

In the recent professional gatherings, numerous CS educators have reported that incorporating computer gaming in programming classes creates high levels of excitement and motivations. Inclusion of gaming components have drastically increased the enrolments and engaged students’ interest in challenging physics and mathematics topics. As eloquently described in the RFP for this proposal, integrating computer gaming into CS1 and CS2, the first programming courses students encounter, is a promising strategy for recruiting and retaining potential students”.

2. Review relevant literature

The literature review is a major component of any research. A literature review allows for a clear understanding of the state-of-the-art of the research area. It allows for the researcher to understand what others have done in attempts to a solution as well as the success rates. The researcher needs a sound understanding of recent literature to avoid wasting time on research problems that have already been addressed.

“Such a review, not only provides exposure to a larger body of knowledge but also equips with enhanced knowledge to efficiently follow the research process. Through a proper review of the literature, the researcher may develop the coherence between the results of his study and those of the others” (IEdunote, 2021)

The importance of reviewing relevant literature is listed below from the IEduNote website.

- It avoids duplication of the work that has been done in the recent past.
- It helps the researcher to find out what others have learned and reported on the problem.
- It helps the researcher to become familiar with the types of methodology followed by others.
- It helps the researcher to understand what concepts and theories are relevant to his area of investigation.
- It helps the researcher to understand if there are any significant controversies, contradictions, and inconsistencies in findings.

- It allows the researcher to understand if there are any unanswered research questions.
- It might help the researcher to develop an analytical framework.
- It will help the researcher to consider the inclusion of variables in his research that he might not otherwise have thought about.

(iedunote, 2021)

3. Draft research questions, objectives, and hypotheses

The research question serves as the centre post for the research. This is the ultimate question that the research sets out to answer.

From the research question, a set of research objectives are derived. These objectives serve as milestones that need to be achieved in order to eventually arrive at an answer to the research question.

Hypothesis is a statement that can be proven to be true or refuted. The hypothesis is developed and then the research sets out to test it.

Example 1:

Research Question

Which is the most effective ML technique for the early prediction of unreliable mobile apps?

Research Objectives

- To identify challenges with current approaches for determining software reliability.
- To determine machine learning algorithms best suited for reliability prediction.
- To model the data using machine learning techniques to evaluate their prediction performance.

Example 2:

Research Question

What is an effective approach for improved business analysis of public eservice systems?

Research Objectives

- Evaluate existing business analysis approaches adopted for public eservices systems projects
- Determine the key knowledge areas, process flows and activities in business analysis applicable to eservice systems
- Develop a framework of practice guidelines for business analysis of eservice systems
- Validate the components and to determine the nature of the interfaces between components in the framework

4. Determine the research strategy

The research strategy is an important decision to be made. The strategy will serve as a framework for the rest of the study. The framework should contain the necessary research components to ensure that the objectives can be achieved.

The research strategy determines how data will be collected, analysed and the guiding philosophy to formulating results.

Types of research strategy include experiment, case study, design research or survey. The choice depends characteristics of the topic such as, the type of problem, the objectives of the study, the state of the data (what form does the data exist, qualitative or quantitative), resources available for the study.

5. Data collection (and sampling)

Data collection can be either from a primary or secondary source. The secondary source includes data that has already been analysed and is available in the form of reports, publications, online data bases and so on. Primary data is data that is collected for the first time in a study.

The main types of data are qualitative and quantitative. Qualitative data is usually collected from people and then analysed for the purpose of research. Quantitative data is usually numerical and transforms participants' responses into numbers using some scale. Qualitative data can be collected via interviews, observations, focus-groups and open-ended questions. Quantitative data is usually collected via surveys, closed questions and questionnaires. When collecting quantitative data, sampling is important.

Sampling is the task of selecting a small number of items (sample) for the research from a much larger group (population). The research is conducted using the smaller sample and the results are generalized to the population. Therefore, it is vitally important that the sample is an accurate representation of the whole population. The opposite of sample is census whereby the entire population is included in the research.

6. Data analysis

Data analysis is the process of careful examination of collected data. The researcher analyses the data searching for patterns and trends, to summarize the data and apply techniques to understand the data and arrive at findings. These are then used to answer research questions, test hypothesis and refute or develop theories.

Data analysis for quantitative data differs to that for qualitative data. Quantitative data analysis utilizes statistical techniques and formula to arrive at results. Data collected in different formats are converted to numbers prior to analysis and arriving at results. Qualitative data is usually coded into a smaller, more manageable format before analysis can begin.

A research study can include a mix of simple data analysis techniques to more complex multivariate techniques. This depends on the research question and the nature of the collected data.

7. Writing up findings

At the end of the research, the findings need to be communicated to the IT research and practitioner community. This comes in the form of conference presentations and proceedings, journal publications and book and book chapters. It is important for the advancement of the field.

2. Components of a research proposal

The easiest way to describe the research proposal is to think of it as the *plan* for the research undertaking. The proposal must provide a detailed plan of how the research will be conducted. The different sections must be logical in their link to each other and flow smoothly from one section to the next.

The proposal is usually peer reviewed and a PhD or Masters student needs to present their proposal to supervisor, co-supervisor (if any) and HOD prior to submission.

The research proposal has to be of good quality. There are limited students allowed into PhD and Masters programs and a limited funding allocation. Hence, the proposal is an important selection tool.

Purpose of the research proposal

The research proposal is a useful instrument for the following reasons:

- To show that the intended research has the potential to make a contribution to the field
- To provide a detailed plan for research design for achieving the objectives
- To show that the student/writer has adequate scientific ability and knowledge to undertake the study
- To show that the research can be achieved with the allocated financial and time constraints

Typical sections of the research proposal

1. Topic/title
2. Context
3. Problem statement
4. Aim
5. Research question/hypothesis
6. Objectives
7. Literature review
8. Research methodology
9. Expected contribution
10. Plan of research activities
11. Budget
12. Structure of thesis (Chapter)
13. Ethics approval section

Guidelines for sections

Title:

Should be concise, descriptive of the study and exciting. The aim should be embodied in the title.

Context of study:

Describe the domain of the research i.e. what area in the literature do you aim to contribute towards?

Introduce the main variables of the study (usually 2 or 3)

Briefly show gaps in literature concerning those variables

Research Problem:

What are the challenges?

What is current literature lacking or gaps in literature?

Main research question

Aims and objectives:

What do you plan to do to solve identified problems?

How will you solve the problems? i.e. state the objectives of the study (3)

Introduce your sub-research questions

Literature review:

Synthesis of literature to make a case for the need for your study.

Research methodology:

Research type, research strategy, research method(s), data collection and data analysis approach, plan to validate findings.

Expected contributions:

Briefly state how your study will contribute towards the body of knowledge

Skills needed to write a good research proposal

The write needs to practiced skills in the following areas in order to develop a high quality proposal.

- Knowledge of the state-of-the-art in the intended area of research. This includes new developments evident in literature and methodology
- Scientific writing and grasp of nomenclature relevant to the field
- Critical thinking to provide a good analysis, evaluation and synthesis of literature
- Communication skills to produce a high quality write-up

Tutorial

1. Write a title and a short problem statement for an IT research study.
2. Write three objectives for the study describe above.
3. Differentiate between experiment, case study, design research and survey research strategies.
4. Differentiate between simple random sampling, stratified random sampling, cluster sampling and systematic sampling.
5. Differentiate between convenience sampling, purposive sampling, snowball sampling and quota sampling.
6. Add one more item to the list for the purpose of the research proposal.
7. Add one more item to the list of skills needed to write a good research proposal.
8. Download the DUT Pg2a research proposal template. Is this a good example of a proposal structure? Provide two points of motivation.

References

Iedunote. 2021. Research Process: 8 Steps in Research Process. Available at: <https://www.iedunote.com/research-process>. Accessed on 06 January 2021.

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